Interface Class

Lecture 22

Based on Slides of Dr. Norazah Yusof

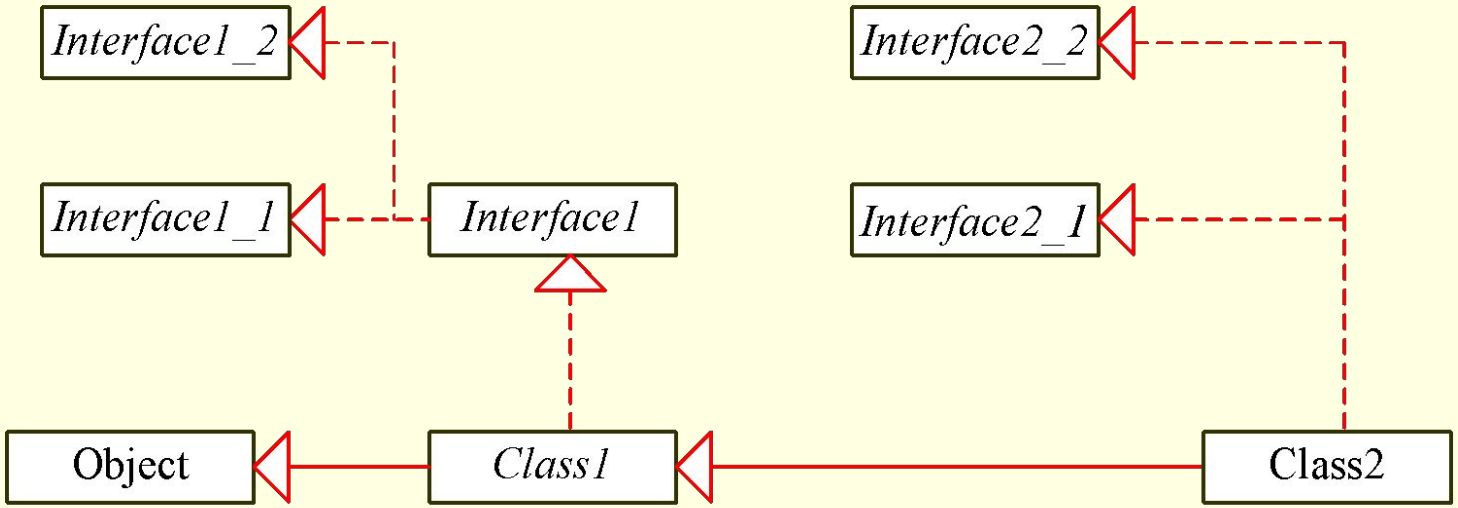
Interface & Implements

- **An interface is a classlike construct that contains only constants variables and abstract methods definition.**
- **An interface cannot contain instance variable and solid methods.**
- **An interface specifies what methods will be implemented by classes that *implements* an interface.**
- **This is another way to design polymorphic methods.**

Interface

- All classes share a single root, the Object class, but there is no single root for interfaces.
- Like a class, an interface also defines a type.
- A variable of an interface type can reference any instance of the class that implements the interface.
- If a class extends an interface, this interface plays the same role as a superclass. You can use an interface as a data type and cast a variable of an interface type to its subclass, and vice versa.

Interface



Suppose that c is an instance of Class2. c is also an instance of Object, Class1, Interface1, Interface1_1, Interface1_2, Interface2_1, and Interface2_2.

Interface declaration

- **Declaration:**

```
public interface InterfaceName
   {
     /* constant declaration */
     /* method declarations (without
   implementations.) */
   }
```

  - **Constant declaration public, static and final**
  - **Method abstract, public**

Interface declaration

In an interface, by default all data fields are constant variables and defined as *public* *final* *static* and all methods are *public* *abstract*.

For this reason, these modifiers can be omitted, as shown below:

```
public interface T1 {
  public static final int K = 1;

  public abstract void p();
}
```

Equivalent

```
public interface T1 {
  int K = 1;

  void p();
}
```

A constant defined in an interface can be accessed using syntax InterfaceName.CONSTANT_NAME (e.g., T1.K).

Ex: Interface declaration (constant variables)

```
public interface Conversionfactors
{
    double inToMm = 25.3;
    double ounToGram = 28.34952;
    double poundToGram = 453.5924 ;
}
```

- **Save as ConversionFactors.java**

Ex: Interface declaration (Abstract Methods)

```
public interface Conversions {
  public double InToMm();
  public double OunceToGram();
  public double PoundToGram();
}
```

- **Save as Conversions.java**

Ex: Interface declaration (variables and Methods)

```
public interface MyInterface    {
    public final int aConstant = 32;
    public final double pi = 3.14159;
    public void methodA(int x);
    double methodB();
}
```

Implementing an Interface

- A class may extends one parent (superclass), but it can implement none or more interfaces
- A class that implements an interface:
  - Can access directly all the variables declared in the interface
  - Have to redefined all the methods declared in the interface
  - Declaration:

```
class SomeClass extends Parent implements SomeInterface
      { ordinary class definition body      }
```

- **The body of the class definition is the same as always.**
- **If any of the abstract methods (in the interface) are not defined in the class that implements the interface, the class will become an abstract class**

Example 1

```
public interface Speakable {
  public String speak();
}

public class Animal {
  protected String kind;
  public Animal() { };
  public String toString(){
   return "I am a " + kind + " and I go " +
((Speakable)this).speak();
 }
}
```

Example 1

```
public class Cat extends Animal implements Speakable {
  public Cat() {
    kind ="cat"; }
  public String speak() {
    return "meow"; }
}

public class Cow extends Animal implements Speakable {
  public Cow() {
    kind ="cow"; }
  public String speak(){
    return "moo"; }
}
```

Interface vs Abstract

- **In many ways, an interface is similar to an abstract class, but**
  - **an abstract class can contain variables and concrete methods as well as constants and abstract methods.**
  - **In an interface, the data must be constants; an abstract class can have all types of data.**
  - **Each method in an interface has only a signature without implementation; an abstract class can have concrete methods.**

Interface vs Abstract

	Variables	Constructors	Methods
Abstract class	**No restrictions**	**Constructors are invoked by subclasses through constructor chaining. An abstract class cannot be instantiated using the new operator.**	**No restrictions.**
Interface	**All variables must be public static final**	**No constructors. An interface cannot be instantiated using the new operator.**	**All methods must be public abstract instance methods**

Comparable Interface

```
// This interface is defined in
// java.lang package
package java.lang;

public interface Comparable {
  public int compareTo(Object o);
}
```

Determine the order of given object with the specified object o, and return a negative integer if this object is less than,
return a zero if this object is equal,
return a positive integer if this object is greater than,
the specified object o.

String and Date Classes

Many classes (e.g., String and Date) in the Java library implement Comparable to define a natural order for the objects. If you examine the source code of these classes, you will see the keyword implements used in the classes, as shown below:

```
public class String extends Object
    implements Comparable {
  // class body omitted
}
```

```
public class Date extends Object
    implements Comparable {
  // class body omitted
}
```

```
new String() instanceof String
new String() instanceof Comparable
new java.util.Date() instanceof java.util.Date
new java.util.Date() instanceof Comparable
```

Generic `max` Method

```
// Max.java: Find a maximum object
public class Max {
  /** Return the maximum of two objects */
  public static Comparable max
      (Comparable o1, Comparable o2) {
    if (o1.compareTo(o2) > 0)
      return o1;
    else
      return o2;
  }
}
```

(a)

```
// Max.java: Find a maximum object
public class Max {
  /** Return the maximum of two objects */
  public static Object max
      (Object o1, Object o2) {
    if (((Comparable)o1).compareTo(o2) > 0)
      return o1;
    else
      return o2;
  }
}
```

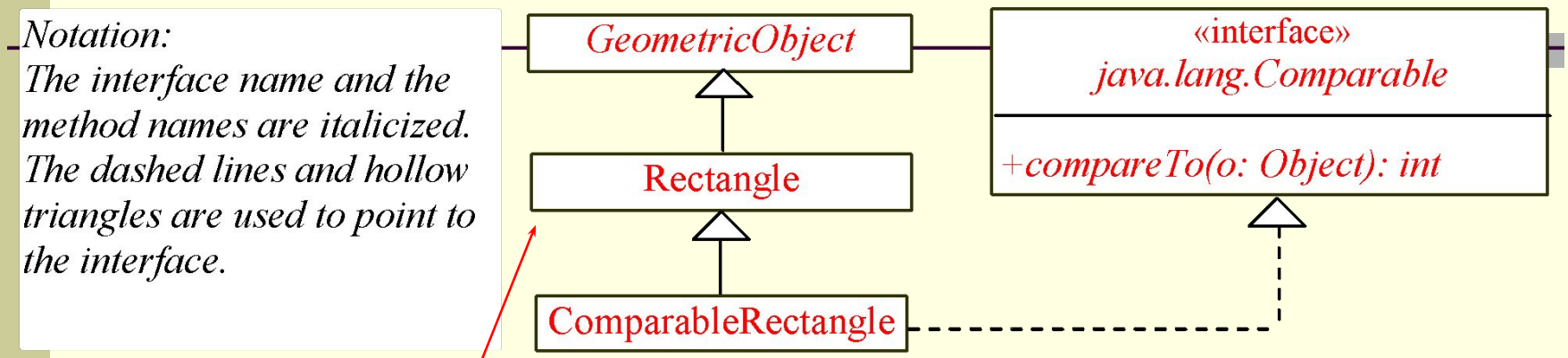
(b)

```
String s1 = "abcdef";
String s2 = "abcdee";
String s3 = (String)Max.max(s1, s2);
```

```
Date d1 = new Date();
Date d2 = new Date();
Date d3 = (Date)Max.max(d1, d2);
```

The <u>return</u> value from the <u>max</u> method is of the <u>Comparable</u> type. So, you need to cast it to <u>String</u> or <u>Date</u> explicitly.

Declaring Classes to Implement Comparable



You cannot use the max method to find the larger of two instances of Rectangle, because Rectangle does not implement Comparable. However, you can declare a new rectangle class that implements Comparable. The instances of this new class are comparable. Let this new class be named ComparableRectangle.

```
ComparableRectangle rectangle1 = new ComparableRectangle(4, 5);
ComparableRectangle rectangle2 = new ComparableRectangle(3, 6);
System.out.println(Max.max(rectangle1, rectangle2));
```

Exercises (Lab 10)

- **Self tests**
  - **Question 3, page 194**
- **Lab Exercises:**
  - **Question 2, page 192**
  - **Question 4, page 194**
  - **Question 1, page 196**